

Machine Learning

Training vs Test Data

Grade 5 | AI & Digital



What is Artificial Intelligence?

AI helps machines think and learn like humans!

- AI is everywhere around us
- It's in your phone and games
- Robots use AI too!



What is Machine Learning?

Machine Learning means teaching computers to learn from data!

- Computers find patterns in information
- They improve by practicing over time
- No one tells them exact answers



Why Do Machines Need Data?

Data is information that helps machines learn.

- More data helps machines learn better
- It's like a recipe — more ingredients, better results!
- Machines need lots of examples

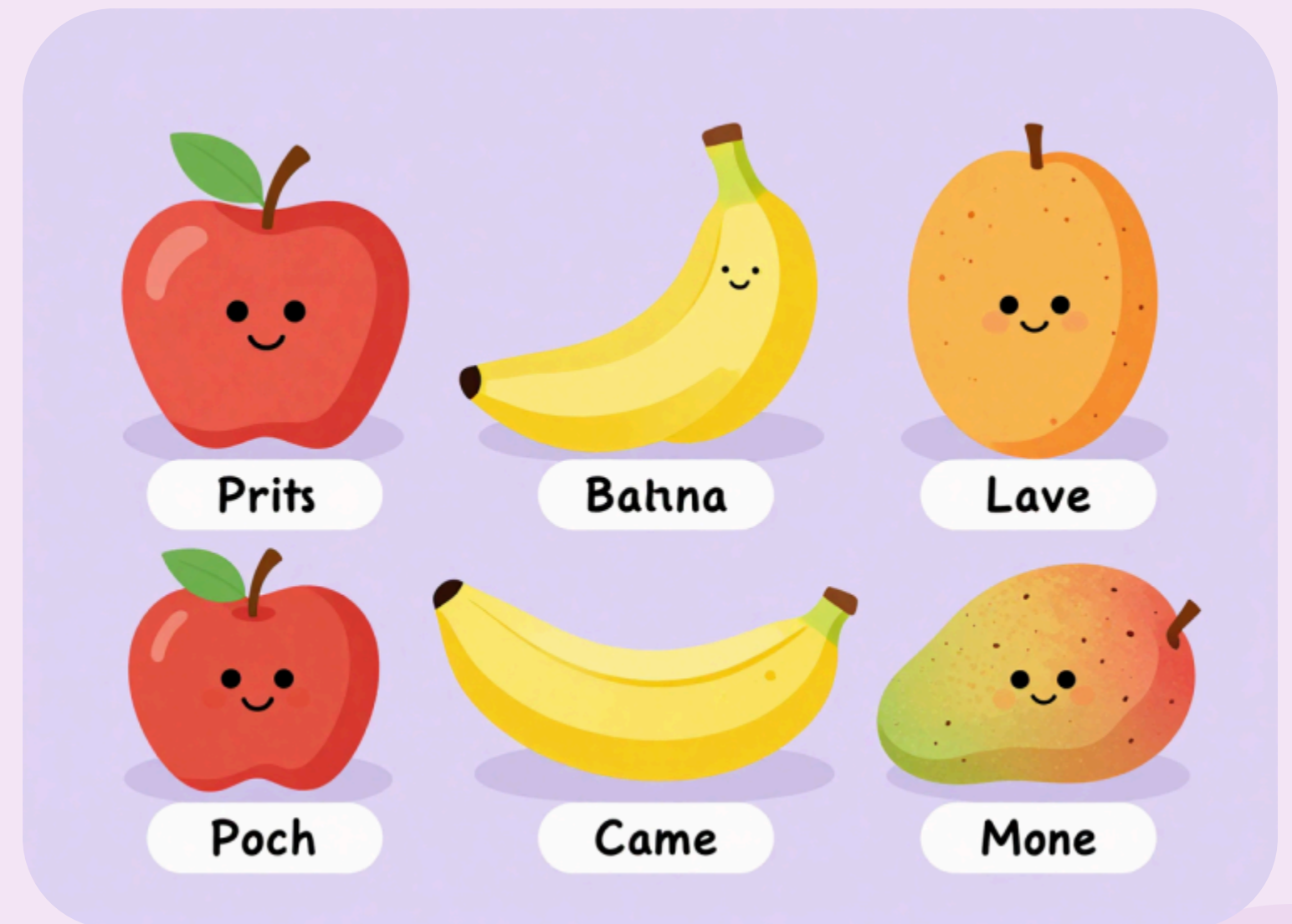


Training Data — Teaching the Machine

Training data is information used to teach the machine.

Example: Showing labeled pictures:

- Apple — 'This is an apple'
- Banana — 'This is a banana'
- Mango — 'This is a mango'



Test Data — Checking the Machine

Test data is new information to check if the machine learned well.

It's like a surprise quiz!

- Show new fruit pictures
- No labels given
- Can the machine guess correctly?



Training Data vs Test Data

Training Data

- ✓ Used to teach machines
- ✓ Has labels or answers
- ✓ Machines learn patterns here
- ✓ Like a textbook with answers

Test Data

- ✓ Used to check learning
- ✓ No labels shown to machine
- ✓ Tests if machine predicts right
- ✓ Like a surprise quiz!

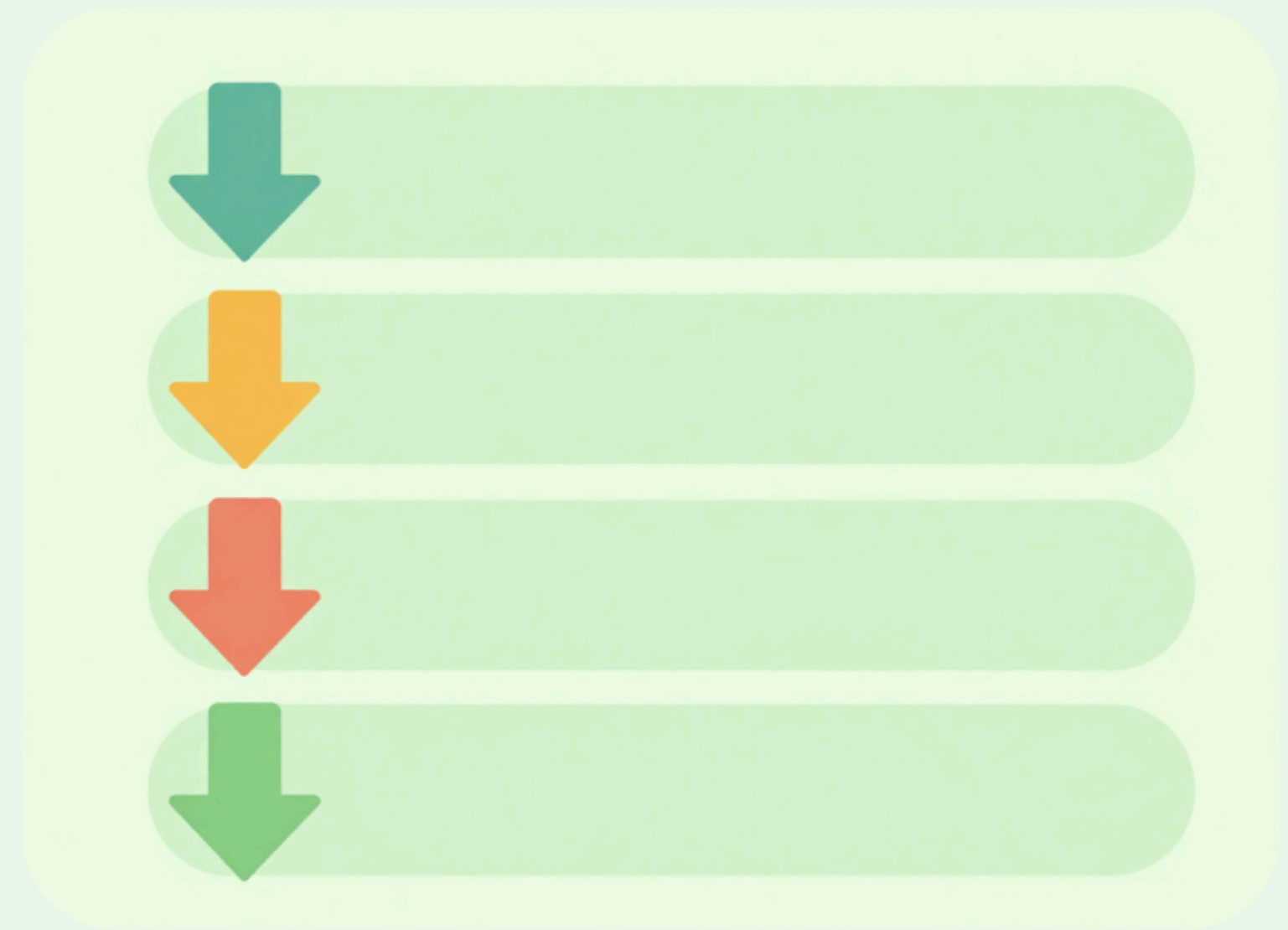
How Do Machines Learn From Data?

Step 1: Look at training data

Step 2: Find patterns and rules

Step 3: Make guesses on new data

Step 4: Check if guesses are correct!

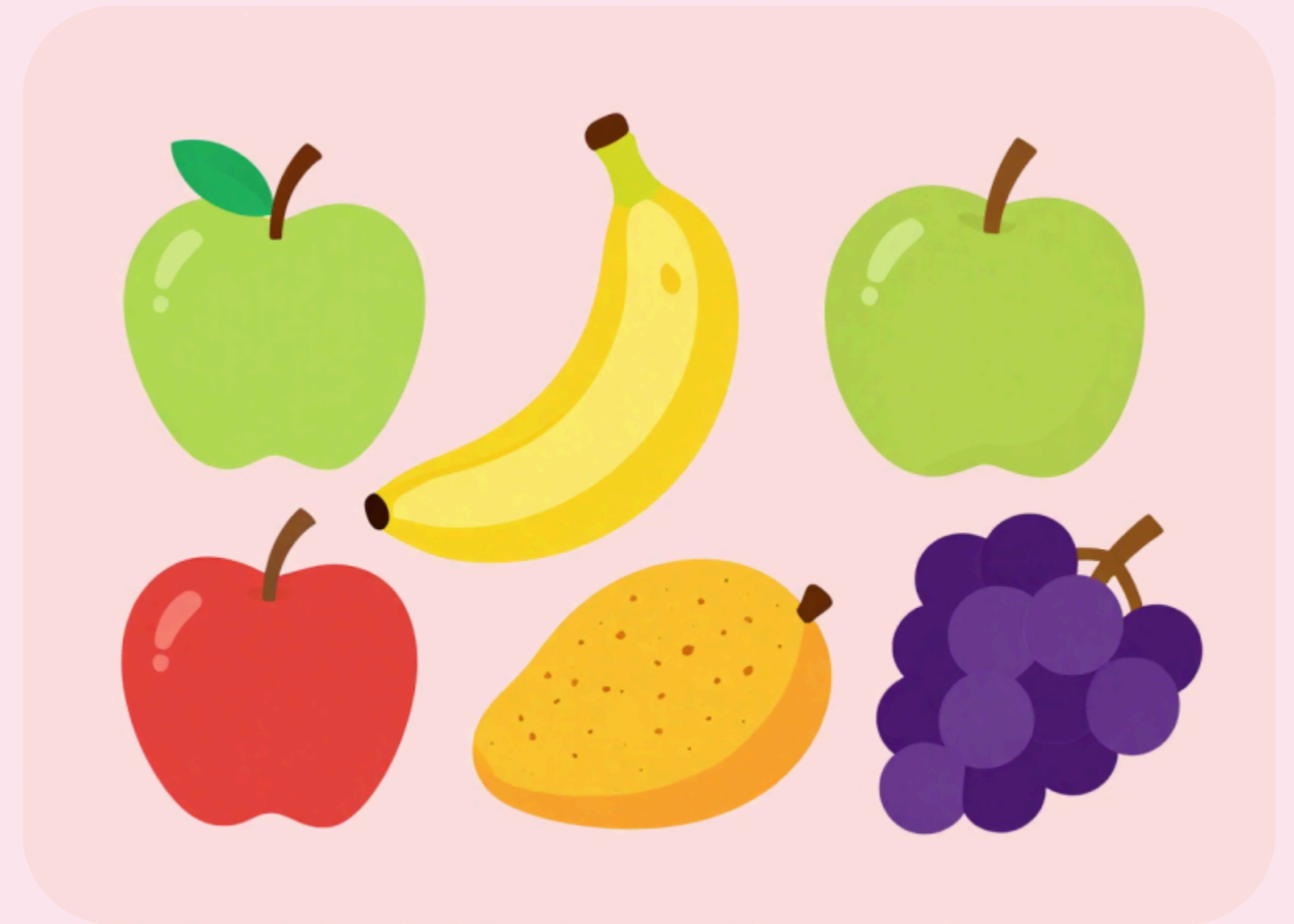


Example:

Recognizing Fruits

The machine learns fruit names by looking at many pictures!

- Training: labeled photos of fruits
- Machine learns: 'Red and round = Apple!'
- Practice makes perfect!



Thank You!

You did an amazing job learning
about Machine Learning today! 🎉

